

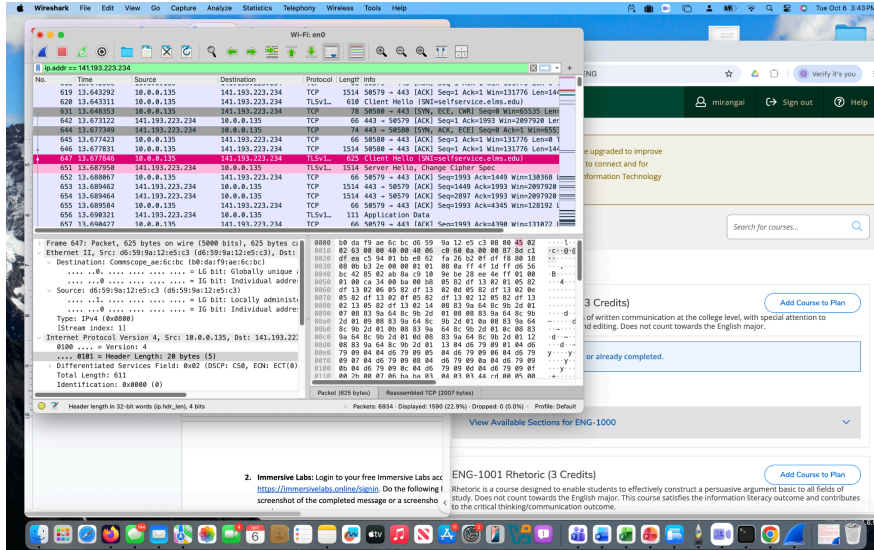
CIT 3405: Digital Forensics

Lab 5: Network Forensics

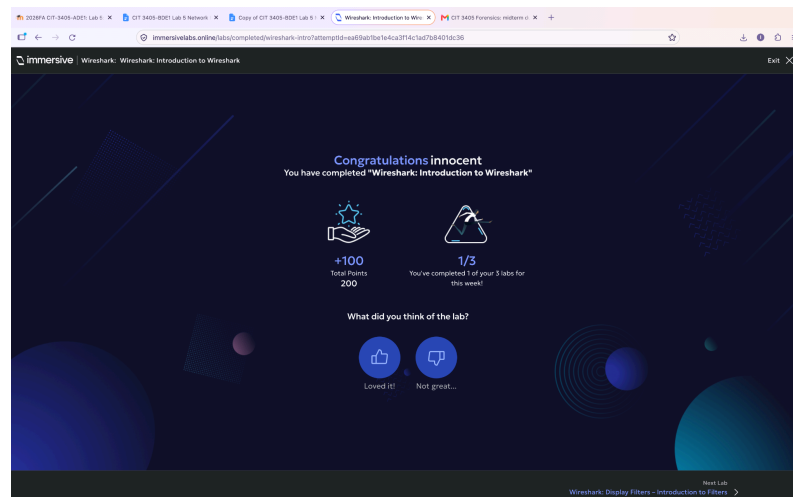
Do File/Make a Copy to make your own editable version of this document. If you get stuck, this [Wireshark Tutorial](#) or the videos on Moodle will help).

1. Wireshark Packet Capture Warm Up:

- a. Download and install Wireshark from <http://wireshark.org> (it will install ncap and wireshark).
- b. Start Wireshark. Click on WiFi (or Ethernet if you're plugged in). See [Wireshark Tutorial](#) for screenshots of how to do this.
- c. Click on the blue shark fin at the top left to start a packet capture session.
- d. Open a browser and go to <https://selfservice.elms.edu/Student/> and login. Click on the Catalog or some other button.
- e. Go back to Wireshark and click on the red square next to the shark fin icon to stop the packet capture.
- f. Type **http** in the filter window at the top (where it says Apply a display filter) and enter to see the http packets. You probably won't see any because this is a secure https website.
- g. Erase what is in the filter window or click on the x all the way to the right of the window. Type **ssl** (secure sockets layer) in the filter window at the top and enter to see the https packets.
- h. Scroll down to find a "**Client Hello SNI=selfservice.elms.edu**" packet followed by a "Server Hello" packet. If you cannot find it, do Ctrl-f and then pull down the filter menu to choose String and search for selfservice.
- i. Look at the Source and Destination columns in the "Client Hello" and the "Server Hello" packets. Your machine is the client. What's your IP address in the source or destination: **_10.0.0.135** (probably starts with 10 or 192).
- j. What is the IP address of selfservice.elms.edu which is the server?
_____141.193.223.234_____ (Hint: it should **not** start with 10 or 192).
- k. Try the filter **ip.addr ==** and then the ip address above to see all the packets with this ip address. You'll see a lot of SYN and ACK handshake packets to establish the connection and encrypted application data packets. See <https://wiki.wireshark.org/DisplayFilters> for more filter examples. Take a screenshot and put it in here. If you couldn't find everything, take a screenshot of what you found and paste it in here.

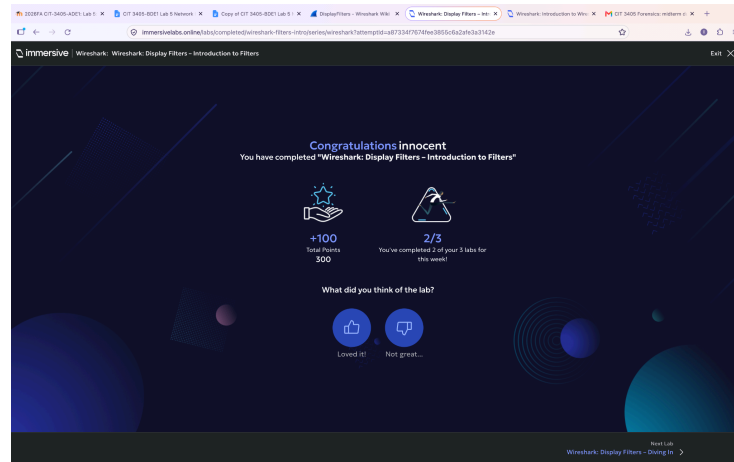


2. **Immersive Labs:** Login to your free Immersive Labs account at <https://immersivelabs.online/signin>. Do the following ImmersiveLabs and capture a screenshot of the completed message or a screenshot of where you got to if you get stuck
 - a. [Wireshark: Introduction to Wireshark](#). Hints: guess on the multiple choice questions, double click on the pcap file in the folder to open it in Wireshark, filter by typing **http** in the filter window and hit enter, scroll down the last http packet after filtering by http, to get the date and time in the right format pull down the **View** menu and click on **Time Display Format** and click on the first choice **Date and Time**, then look at the Time column of the last packet and try to exactly match the given pattern they want.



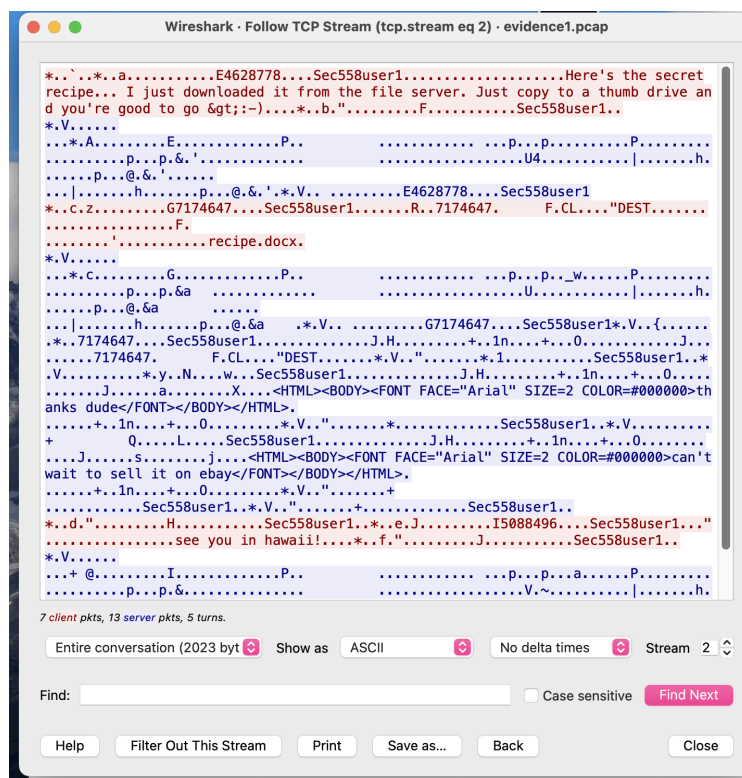
- b. [Wireshark Display Filters - Introduction to Filters](#). Hints: Filter by typing **tcp**. The number and percentage of packets after filtering is shown in the status bar at the very bottom of the Wireshark window but it gets blocked or cut off (try to move

the top of the window up to see it). Use filters with ip.src and ip.dst and tcp.port (<https://wiki.wireshark.org/DisplayFilters>).



3. **Network Forensics Case:** In this case study, you will be investigating an employee who is suspected of passing company secrets to a competitor. The employee's computer has the IP address **192.168.1.158**, and the company has obtained 2 packet capture files around the time of the suspected crime.
 - a. Download the [evidence1.pcap](#) (packet capture file) to your own computer.
 - b. Open evidence1 by double clicking or open Wireshark on your own computer and do File/open. If the top window is small, drag the bottom edge down to resize it.
 - c. What is the non-company IP address that the employee at **192.168.1.158** was chatting with? You may want to filter by **ip.addr == 192.168.1.158**. Look for TCP or SSL packets sent or received by an IP address that is not a company computer (doesn't start with 192.168). What IP address did you find? **64.12.24.50**
 - d. Are the chat messages incriminating? Right click on the TCP or SSL packets and select "Follow TCP Stream" to view the packets of the chat conversation. Paste a

screenshot of the chat messages found in the packets here.



- e. One of the messages mentions a filename that might be for the secret recipe. What's the filename? recipe.docx.

4. **Network Miner:** We can recover the transmitted files in two ways by using file carving the Follow TCP Stream or by using a tool like Network Miner. Download [Network Miner](#) on a Windows computer (come to the CIT lab if you need one) and extract the files and **right click** on the executable NetworkMiner.exe to **run as Administrator**. Do File/Open to open the evidence1.pcap file that you downloaded above in Network Miner. Click on Files and then right click on a file to open it. Paste in the recipe you found here. Note that you can also read the messages better in this tool in the Messages tab. (For an extra optional challenge, you can try file carving from Wireshark. Click on Save As in the Follow TCP stream window in Wireshark and click on the Save As button and save in any file and then open the file in HxD and find the right [file signature](#) to carve the file).
5. Continuing on with the story, the employee who stole the secret recipe is Ann Dercover and she has left the country! You have a new packet capture file [evidence2.pcap](#) of a couple of her SMTP email messages before she left. Open **evidence2.pcap** in **Network Miner** (not Wireshark although it may work too). Find the following looking in the Messages, Credentials, and Files tabs:
- a. Ann's email address: sneakgy33@aol.com

- b. The 2 email addresses that she was corresponding with:
 ___**sec558@gmail.com**_____
 - c. Her email username and password (this was back before emails were encrypted in 2010): __**username: snakyg33k@aol.com**
 password:558r00lz_____
 - d. Where she went. This is in an attached file (right click to open file). Where did she go? __**lunch next week**_____
6. Click on Share, change to Anyone or Elms can view, and copy the link to this document and submit the link on Moodle.